

Metal complexes of amino acids and amino acid side chain groups. Structures and properties†

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α -Amino acid residues exhibit various functions in biological systems. The side chain groups are very important for formation of the metal center and its catalytic action in proteins. They are involved in metal binding, molecular recognition by weak interactions, enzyme catalysis, and formation of the molecular environment. This perspective focuses on recent studies on characterization of metal–amino acid complexes involving imidazole, phenol, indole, or alkyl groups, their reactivities, and some selected metal–side chain group interactions.

Introduction

Proteins are constructed from 21 different α -amino acids including selenocysteine as the smallest molecular units. The amino acids form stable five-membered chelates with various metal ions through the amine and carboxylate moieties (*N,O*-chelation).^{1–3} Several amino acids have an additional metal binding site in the side chain and thus form metal complexes with a variety of structures. Groups such as the imidazole ring of histidine (His), the phenol ring of tyrosine (Tyr), and the thiol group of cysteine (Cys) are important metal binding sites in proteins, and in addition

the β - and γ -carboxylate groups of aspartate (Asp) and glutamate (Glu), respectively, and the thioether moiety of methionine (Met) are often involved in metal binding.^{2,4} These and other amino acid side chain groups form a microenvironment around the metal center, which is necessary for recognition and fixation of substrates and catalytic functions of enzymes by noncovalent or weak interactions such as hydrogen bonds, π – π stacking interactions, and hydrophobic interactions.⁵ Weak interactions are important also for chemical systems such as enantioselective Diels–Alder reactions catalyzed by Cu(II) *etc.* in aqueous solution where stacking interactions are effective.⁶ Some side chain groups function as enzyme co-factors represented by the Cu(II)-coordinated Tyr phenoxyl radical enabling two-electron oxidation in galactose oxidase^{7–10} and a built-in co-factor, trihydroxyphenylalanine (topa) quinone, formed at the metal center of copper amine oxidase by post-translational modification,^{9,11,12} and further possibilities may be expected for various other side chain groups.

Fundamental aspects of metal–amino acid complex formation have been well established,^{1–4} and it will not be necessary to treat

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† This paper is dedicated to Prof. Dr Helmut Sigel on the occasion of his 70th birthday with the very best wishes for his health and future activities.



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them in detail. In our previous perspective¹³ we focused on weak interactions involving the side chain polar, charged, and aromatic groups, and showed their effects on the spectral and thermodynamic properties and the structures of ternary complexes with ligand-ligand interactions. Interesting findings have since been reported for metal-amino acid and related systems. Apart from metal complexes in biological systems, combination of metal coordination and weak interactions gives rise to molecular assemblies showing unique structures and functions.^{14,15} The imidazolate-bridged dinuclear structure revealed for CuZn-superoxide dismutase (CuZn-SOD)¹⁶ was realized in metal complexes¹⁷ and is now widely recognized as an important structural unit for molecular architecture.^{14,15,18}

With these points in mind, we intend to give a perspective of some recent advances in the studies on metal-amino acid and related complexes and activities of the amino acid side chain groups. For this purpose we selected papers from the point of our current interest with emphasis on the metal binding and noncovalent interactions of imidazole, phenol, indole, and alkane moieties contained in complexes and their possible biological relevance.

Complex formation of amino acids with and without the side chain coordinating groups

Metal ions bind with simple α -amino acids (aa) by *N,O*-chelation, forming a five-membered chelate ring.^{1–3} Amino acids are known to form very weak complexes with alkali metal ions and weak complexes with Mg(II) and Ca(II).^{19,20} but the complex species and their structures in solution have been elusive due to low stability. Fox *et al.* recently studied the crystal structures of Ca(II) complexes of several neutral amino acids and revealed the existence of a mononuclear complex of alanine (Ala) with a 2N5O donor set, [Ca(Ala)₂(H₂O)₃], and one-dimensional coordination polymers of the other amino acids.²¹ It is interesting to note

that in all of the complexes aa binds with Ca(II) by *N,O*-chelation, and this may support that this binding mode prevails also in solution. Infrared multiple photon dissociation (IRMPD) spectroscopic studies were reported for the structures of the Ba(II) complexes of arginine (Arg), glutamine, *etc.*²² Structural characterization of the complexes of amino acid derivatives has been made for the vanadyl complexes of Schiff bases derived from 2-naphthaldehyde and Tyr, serine (Ser), or threonine (Thr)²³ and the dinuclear Cu(II) complexes of glycine (Gly), Ala, and phenylalanine (Phe) tethered to triazacyclononane through an amide bond, where the amino acid moiety binds with the metal ion by *N,O*-chelation.²⁴ Structures and stabilities of lanthanide-amino acid and peptide complexes have been reviewed.²⁵ Recently, there is an increasing interest in bioorganometallic chemistry of ferrocene analogs containing amino acids. Amino acids were found to coordinate to bent metallocenes such as molybdocene²⁶ and vanadocene²⁷ by *N,O*-chelation, but cysteine was revealed to bind with vanadocene by *O,S*- and *N,S*-chelations at acidic and neutral pH, respectively.²⁸ Complex formation of organometallic compounds with amino acids, peptides, *etc.*, has been reviewed.²⁹

Speciation studies were made for Al(III)-aa systems (aa = Gly, Ser, Thr, His) under physiological conditions (*I* = 0.15 M (NaCl) and 37 °C), which gave a doubly deprotonated species, M₂LH₂ (L, free aa; H, proton), as the only significant species.³⁰ Theoretical calculations were carried out on complexation of Al(III) with the neutral form of aromatic amino acids (Phe, Tyr, and tryptophan (Trp)) in the gas phase;³¹ in the most stable structure Al(III) binds with the amine N and carboxylate O atoms with the aromatic ring located above the coordination plane and in a cation- π interaction.³² In the Trp complex it forms a covalent bond with one of the C atoms of the pyrrole moiety of the indole ring ($d_{\text{Al-C}} = 2.04 \text{ \AA}$), which indicates that the aromatic amino acids behave as tridentate ligands (*vide infra*).³¹ The stability of the complexes was found to be in the order Trp > Tyr > Phe due to the polarizability arising from the ring, *i.e.*, the higher the polarizability, the more stable the complex. X-Ray structural and solid-state NMR studies were reported for Pb(II)-aa complexes (aa = neutral isoleucine and valine (Val) (species LH), deprotonated leucine (L), and monodeprotonated Asp (LH)), showing that aa binds with Pb(II) through the carboxylate oxygen atom while deprotonated leucine binds with Pb(II) by *N,O*-chelation.³³

Among the side chain coordinating groups, the imidazole ring of His and the β -SH moiety of Cys, which are important and frequently observed metal binding sites in proteins, often replace the carboxylate group in metal-His and Cys complexes, resulting in *N,N*- and *N,S*-chelations, respectively. Cu(II)-His complex formation has long been studied,^{34–36} and recently a number of studies have been reported for solution equilibria of metal-His and His-containing peptide complexes (metal (M) = Cu(II), Ni(II), Zn(II)) in relation to the metal binding by the octarepeat region of prion proteins and amyloid- β .³⁷ The side chain involvement in metal binding by oligopeptides has also been reviewed and discussed with emphasis on the His imidazole and Cys thiolate moieties.³⁸ A ¹H NMR study was made on the Cu(II) binding to His-containing peptides related to the prion protein by using diamagnetic replacement ions such as Pd(II).³⁹ There can still be various modes of coordination of His and His-containing peptides depending on pH and the presence of other ligands. The bonding modes and structures of Cu(II)-His complexes in aqueous



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solution (pH 0–10) were recently studied by various spectroscopic methods including IR and Raman spectroscopies, and formations of different, partly coexisting complex species were proposed at various pH values ranging from 0 to 10,⁴⁰ which confirmed or refined the structures reported in the literature. By using ESR parameters obtained by the two-dimensional computer simulation method, characterization of the binding modes in the Cu(II)–HisGly system (HisGly = histidylglycine) was performed for the coexisting species, which are often structurally indistinguishable by pH titration alone.⁴¹ The unique structure of [Cu(His)₂] (1), which has been used for treatment of Menkes disease, was recently determined by Sarkar *et al.* as shown in Fig. 1,^{36,42a} where one of the imidazole rings remains uncoordinated in contrast to the meso complex [Cu(L-His)(D-His)(H₂O)₂] having both imidazole rings coordinated.^{42b}

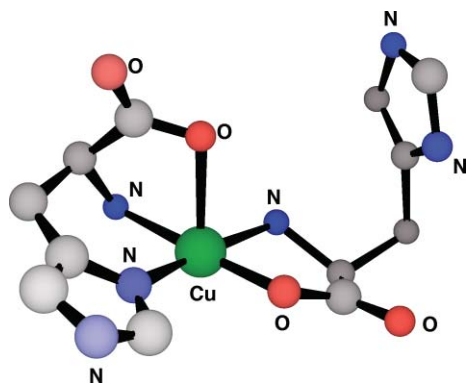


Fig. 1 Structure of [Cu(His)₂] (1).^{42a}

Coordination of the side chain carboxylate group of Asp and Glu and the thioether group of methionine (Met) depends on the stereochemistry and the hard and soft properties of the metal ions. For example, the Asp moiety of salicylaspartate binds with VO(II) through the deprotonated amide nitrogen and the two carboxylate groups at neutral–basic pH.⁴³ Met binds with intermediate metal ions by *N,O*-chelation and with soft metal ions by *N,S*-chelation^{4,44} as seen in the structures of [PdCl₂(DL-Met)] (2)⁴⁵ and [PtCl₂(L-Met)] (3)⁴⁶. The structures of the stereoisomers arising from the side chain group coordination of Met to Pt(II) were studied by X-ray analysis, NMR spectroscopy, *etc.*⁴⁷ The thioether sulfur of Met is well established as a donor atom in blue copper proteins with a distorted tetrahedral coordination geometry,⁴⁸ and methionylglycine binds with Pt(II) through the amino nitrogen and thioether sulfur atoms,⁴⁹ while glycylmethionine binds with Pt(II) through the amine and peptide nitrogen atoms and the sulfur atom.⁴⁶ A mixture of Cu(NO₃)₂·2H₂O, AgNO₃, and L-Met in an ethanol/water diffusion system gave a helical assembly, [(Ag₃Cu₃(L-Met)₆(NO₃)₃(H₂O)₃)·7H₂O]_n (4), by a self-assembly process as a result of the soft-hard recognition and polymerization by Ag–S bridges (Fig. 2).⁵⁰ It is interesting to note that Met with a ferrocene substituent at the amine nitrogen forms a bis-complex with Cu(II) by *N,O*-chelation and additional binding by the thioether sulfur at one of the axial positions with the Cu–S distance of 2.791 Å, which is shorter than that in plastocyanin (2.9 Å).⁵¹

On the other hand, the ε-amino group of lysine (Lys) has a high p*K*_a value (10.71)⁵² and is protonated in proteins. In metal-Lys complexes, the ε-amino group is not involved in metal binding

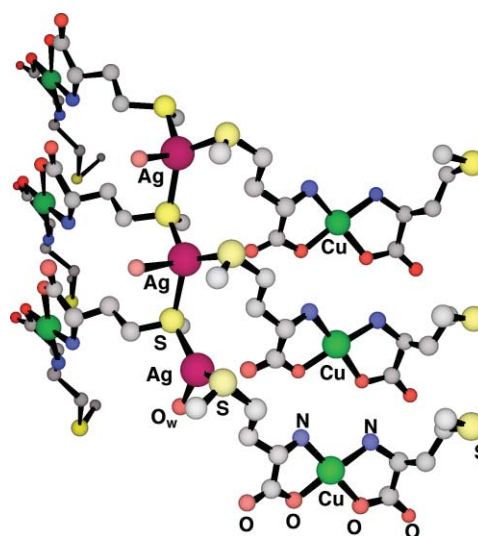


Fig. 2 A helical assembled complex 4 composed of *N,O*-chelated Cu(L-Met)₂ and Ag(I) bound to the thioether sulfur atom.⁵⁰

for steric reasons and is protonated at neutral pH, and even at higher pH where the ε-amino group is deprotonated to give an anionic form of Lys, it remains uncoordinated.⁴ The Cu(II) complex of lysyltyrosine, however, has the 3N1O-donor set in the plane with the α-amino and peptide nitrogen atoms, the carboxylate oxygen, and the ε-amino nitrogen from a neighbouring complex, showing that the ε-amino group coordinates to Cu(II).⁵³ The guanidine moiety of Arg has an even higher p*K*_a value (12.09),⁵² being protonated in a wide pH range and thus not involved in metal binding under usual conditions. The guanidinium group in proteins is known to serve as a hydrogen bonding partner and fix negatively charged substrates such as a peptide carboxylate group in carboxypeptidase A.⁵⁴ It was shown earlier that [PtCl(terpy)]⁺ (terpy = 2,2′:6′,2′′-terpyridine) reacted with cytochromes *c* from horse and tuna upon incubation at pH 7.0 at the guanidine moiety of Arg 91 in addition to the His imidazoles.⁵⁵ The binding mode of the guanidine moiety was established by X-ray analysis, for example, for canavanine, the δ-oxa analog of Arg, in [(Pt(terpy))₂(canavanine)]⁺ (5)⁵⁶ and *N,N*-dimethylguanidine in [Pt(dien)(*N,N*-dimethylguanidine)]²⁺ (6) (dien = diethylenetriamine),⁵⁷ where the guanidine moiety was planar. Arg 260 of biotin synthase was later found to be located close to the Fe₂S₂ cluster as shown in Fig. 3, which indicates that one of the nitrogen atoms of the guanidine moiety is coordinated

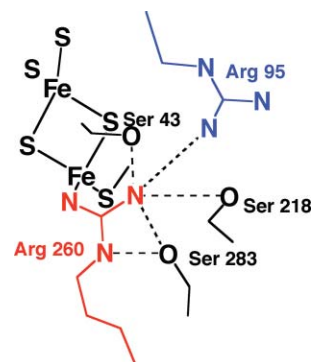


Fig. 3 Fe₂S₂ cluster–Arg 260 binding in biotin synthase.⁵⁸

to the Fe atom.⁵⁸ This is the first observation of the guanidine involvement in the interaction with a metal ion in proteins.

When Tyr binds with a metal ion by *N,O*-chelation, the phenol OH group is not involved in coordination for steric reasons in contrast to the metal binding in proteins, but Tyr-containing peptides such as tyrosylglycine (TyrGly) bind with Cu(II) through the amino and deprotonated peptide nitrogen atoms and the carboxylate oxygen and form a dimeric complex by the deprotonated phenol moieties, which connect two Cu(TyrGly) complexes at pH 8–11 (Fig. 4).^{59,60} The indole ring in Pd(II) and Pt(II) complexes has been shown to bind with the metal ions through the nitrogen and carbon atoms by cyclopalladation.⁶¹ In this connection the methyl ester of Phe was reported to react with Pd(II) acetate in acetonitrile to give a complex formed by *N,C*-chelation involving the C(2) atom of the benzene ring, although free Phe gave a complex by *N,O*-chelation.⁶² In the Cu(I) complex of Me₂IEP containing a pendent indole ring, [Cu(Me₂IEP)]⁺ (7), the indole C(2)–C(3) moiety interacts with Cu(I) to form a bond with distances of 2.228(5) and 2.270(5) Å between Cu(I) and the C(2) and C(3) atoms, respectively (Fig. 5(A)) (*vide infra*).⁶³ Very recently a periplasmic protein, CusF, which is thought to be a copper chaperone or regulator, was revealed by X-ray analysis to have the Cu(I) ion bound to the two Met sulfur atoms and an imidazole nitrogen atom and in addition in the bonding interaction with the indole C(4)–C(5) moiety of a proximal Trp residue with the Cu(I)–C distances of 2.67 and 2.86 Å (Fig. 5(B)).⁶⁴ These Cu(I)–C distances are longer than those of 7 shown in Fig. 5(A), and the interaction was described as a cation– π interaction but may be an interaction lying between a covalent bond and a purely electrostatic interaction.⁶⁵ This is the first example of the metal–indole direct interaction in biological systems. Cation– π interactions involving the indole ring have been recognized in Cu(II)–GlyTrp (GlyTrp = glycyltryptophan) between Cu(II) and C(4) of the indole ring,⁶⁶ and an ESR study was made on the role of such interactions in magnetic exchange interactions between Cu(II) ions.⁶⁷ The fluorescence enhancement in the Ca(II)–EDTA–(Trp)₂ system, which has two Trp residues tethered to the carboxylate moieties of EDTA by amide bonds, was attributed to the Ca(II) $\cdots$ indole contact at the C(3)–C(4) moiety,⁶⁸ and alkali metal– π interactions were shown to occur at the pyrrole ring of indole.⁶⁹

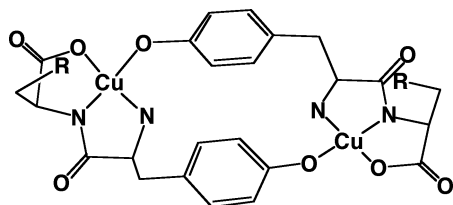
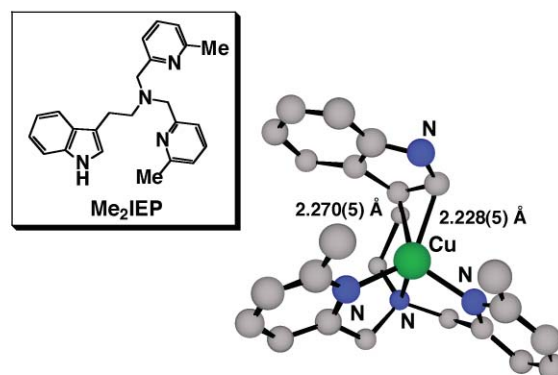


Fig. 4 Phenolate bridges in the dimeric Cu(II)–TyrGly complex (R = H).^{59,60}

Weak interactions involving the amino acid side chain groups

Weak or non-covalent interactions of the amino acid side chain groups are important for structures and properties of proteins⁵ and low-molecular-weight complexes,^{70,71} where charged groups may be involved in electrostatic interactions while aromatic rings

(A)



(B)

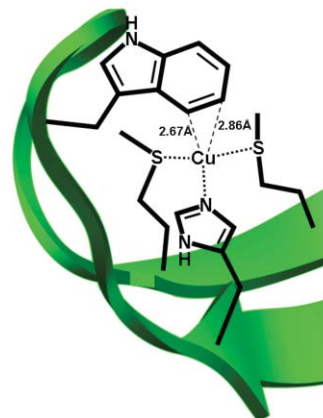


Fig. 5 Cu(I)–indole interactions: (A) Cu(I)–indole η^2 -bond in [Cu(Me₂IEP)]⁺ (7)⁶³ and (B) structure of a periplasmic protein CusF.⁶⁴

may undergo π – π ,⁷² CH– π ,⁷³ cation– π ,³² and d– π ⁷⁴ interactions. Molecular recognition patterns as seen in ternary metal complexes involving amino acids, peptides, and nucleobases/nucleotides were recently reviewed on the basis of the complex stability and ligand–ligand interactions.⁷⁵ In the plastocyanin from a fern, *Dryopteris crassirhizoma*, aromatic ring stacking was observed between the imidazole ring of one of the coordinated His residues, His 90, and the phenyl ring of Phe 12,⁷⁶ and at the Cu site of galactose oxidase the equatorially coordinated Tyr phenolate moiety stacks with the indole ring of a nearby Trp residue, Trp 290, which was shown to play critical roles in stabilization of the phenoxyl radical formed in the course of the enzymatic reaction.^{10,77} Trp residues have also been found to be close to the metal center in proteins such as cytochrome *c* peroxidase, where the indole rings from Trp 191 and Trp 51 stack with the coordinated imidazole ring and the heme porphyrin ring, respectively,⁷⁸ and the indole ring of Trp 191 is converted to the radical in the course of the reaction (*vide infra*).⁷⁹ Stacking and hydrogen bonding modulate the properties of the metal center. As expected from the finding on the fern plastocyanin,⁷⁶ mutants of a pseudoazurin having an aromatic amino acid (AA) in place of Met 16 located close to the Cu site in the wild type exhibited a higher redox potential probably due to the stacking interaction between the aromatic ring of AA and the coordinated imidazole of His 81,⁸⁰ which is in line with the higher value observed for the fern plastocyanin as compared with poplar and other higher plant plastocyanins,⁷⁶ and may be

attributed to delocalization of electrons over stacked aromatic rings. It was noticed that the effect of mutation depends on the subtle difference in the mode of π - π stacking in the mutants.^{80,81}

Flavin mononucleotide (FMN) is the prosthetic group of flavodoxins, which are one-electron carriers belonging to the flavoprotein family. The isoalloxazine moiety of FMN in *Anabena* flavodoxin is sandwiched by the aromatic rings of Tyr and Trp (Fig. 6(A)).⁸² Depending on the sources of flavodoxins, FMN exhibits different redox potentials, which indicates that the π - π stacking and the electrostatic environment of the FMN binding site modulate the redox properties of the co-factor.^{83,84} Ternary Pt(II) complexes involving 1,10-phenanthroline (phen) and an amino acid (Gly, Ala, Arg, or Val) were shown by ¹H NMR to form an adduct with FMN, whose mode of interaction is inferred to be similar that of structurally related anthraquinone-5-sulfonate (AQS), [Pt(phen)(Ala)](AQS) (**8**), revealed by X-ray analysis (Fig. 6(B)).⁸⁵ The adduct formation caused an anodic shift of the potential, $E_{1/2}$, for the two-electron redox process of FMN, which increased with the concentration of the Pt(II) complex and may be due to the lowered electron density by stacking with the electron-deficient Pt(phen) moiety and delocalization of electrons. Redox-tuning effects have been known for the NH...S hydrogen bond between the peptide NH and coordinated Cys S⁻ moieties and studied with model complexes of iron-sulfur proteins, molybdenum enzymes, *etc.* The existence of single and double

NH...S hydrogen bonds were shown for the Fe complexes of Z-Cys-Gly-AlaOMe (Z = benzyloxycarbonyl) and *o*-(acylamino)- and 2,6-(diacylamino)benzenethiolates such as [Fe₄S₄(Z-Cys-Gly-AlaOMe)₄]²⁻ as ferredoxin and rubredoxin models,^{86,87} and similar hydrogen bonds were also established for Mo(IV and V), Zn(II), Cd(II), and Hg(II) complexes of benzene(mono or 1,2-di)thiolate derivatives.^{88,89} The shift of the redox potential to a higher value was observed in these complexes. The NH moiety of the indole ring which is very weakly acidic (for Trp $pK_a = 16.82$)⁹⁰ is involved in hydrogen bonding with the carboxylate group of Asp 235 in cytochrome *c* peroxidase⁷⁸ and in anion binding as seen in the sulfate-binding protein,⁹¹ and other examples of anion binding are known for artificial systems.⁹² It was also shown to be hydrogen-bonded to the axially coordinated water molecule in the Cu(II) complex of an octapeptide mimicking the octarepeat region of the prion protein.⁹³

Metal complexes as hydrolytic metalloenzyme models have long been studied.^{94,95} Regiospecific and stereospecific binding of amino acids was shown by the reaction of a racemic mixture of a chiral Co(III) complex with DL-Ala; the X-ray and NMR studies of a diastereomeric mixture of the complexes shown in Fig. 7 indicated that the steric requirements for the α -H atom of coordinated Ala are different with regard to the interaction with a ligand methyl group, favouring the D-enantiomer with the H atom exposed and thus more easily deuterated in D₂O.⁹⁶ The Zn(II) complex of a tripod-like ligand with a pendent Phe residue bound through an amide bond was found to have a trigonal bipyramidal structure with the side chain aromatic ring near to Zn(II) with the distance of 4.5 Å through the open site in the trigonal plane, suggesting possible applications in enantioselective molecular recognition and hydrolysis.⁹⁷ In the corresponding Cu(II) complex the side chain benzyl group was not in contact with the Cu center in a square-pyramidal geometry with the axial coordination of the side chain amide oxygen atom.⁹⁸ Kinetic resolution of DL-Phe was attained by stereoselective incorporation into the Co(III) complex of a chiral ligand, *N*-carboxymethyl-*N*-pyridylethyl-L-leucine,⁹⁹ the ternary complex crystallized in two forms, one of which exhibited a π - π stacking interaction between the benzene ring of L-Phe and the coordinated pyridine ring while the other containing D-Phe had no such interaction. Formation of the L-Phe-containing complex was 6 times faster than that of the D-Phe-containing one, enabling the kinetic resolution of DL-Phe. In the ternary Co(III) complex containing bis-*N,N*-carboxymethyl-L-phenylalanine and a dipeptide with a bulky side chain at the C-terminal, glycylphenylalanine (GlyPhe), multi-site

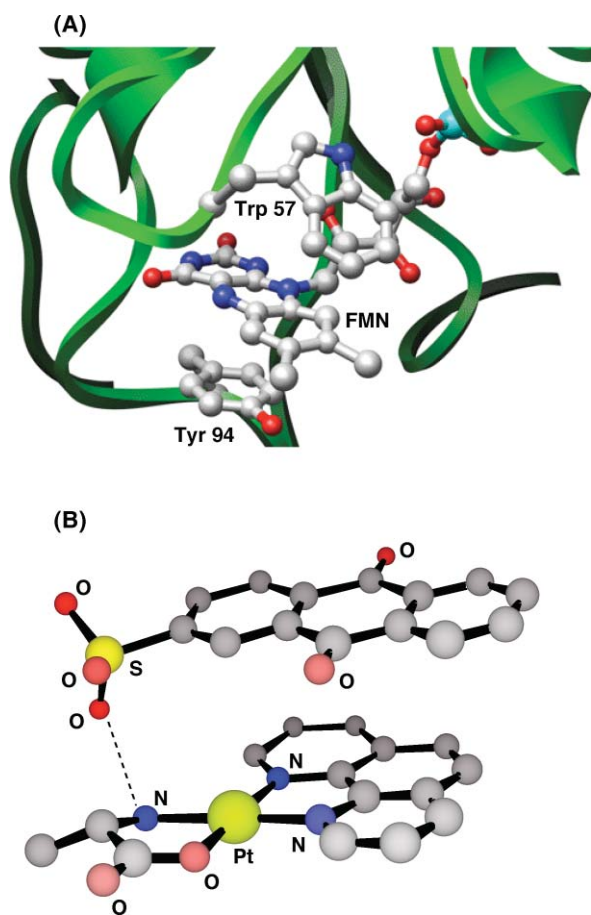


Fig. 6 (A) Active site structure of oxidized flavodoxin from *Anabaena* 7120 with bound co-factor FMN.⁸² (B) Structure of ternary Pt complex-AQS adduct **8**.⁸⁵

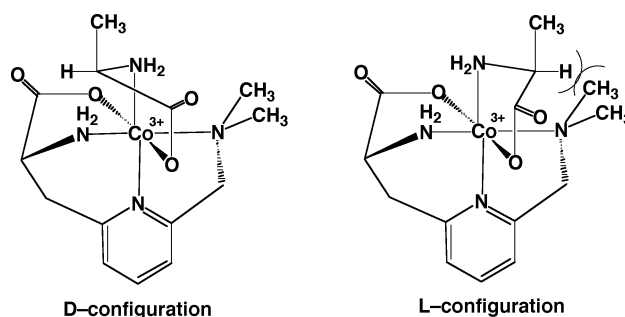


Fig. 7 Structures of chiral Co(III) complexes incorporating D- and L-Ala.⁹⁶

ligand–ligand interactions including a CH– π interaction of the benzene ring of GlyPhe took place and completely inhibited the peptide hydrolysis (Fig. 8).¹⁰⁰

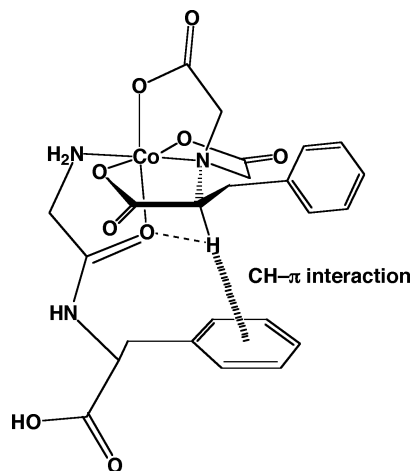


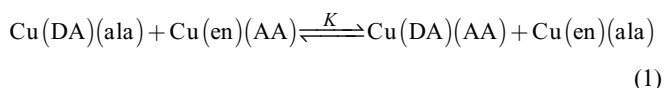
Fig. 8 Weak interactions in a ternary Co(III) complex containing GlyPhe.¹⁰⁰

Zarić *et al.* analyzed the crystal structures of metal complexes available from the Cambridge Structural Database for the interaction of the hydrogen atoms of metal-coordinated ligands with aromatic π systems, which is called a metal ligand aromatic cation– π interaction (MLAC π) and is a special type of cation– π interaction.^{101,102} For example, the interaction involves a hydrogen atom of a coordinated amino group pointing towards an aromatic π system and may be regarded as an X–H $\cdots$ π bond. In the case of alcohol dehydrogenase the methylene group of Zn(II)-coordinated substrate ethanol is located close to the centroid of the phenyl ring of Phe 93 with the distance of 3.19 Å.¹⁰² The π -system of the planar chelate ring of metal–acetylacetonate with delocalized bonds acts as a hydrogen atom donor (metal–ligand CH/ π) or a hydrogen atom acceptor (CH/ π),¹⁰³ and in the latter case the energy of interaction of the Ni(II) complex with benzene was calculated to be similar to that calculated for the T-shaped benzene dimer.¹⁰⁴

Interactions between the metal center and the side chain aromatic and alkyl groups

Intramolecular aromatic ring stacking is well established for ternary metal complexes involving an aromatic diimine ligand (DA) and AA, M(DA)(AA). Their structures and spectral and thermodynamic properties have been described, and stabilization of ternary complexes due to stacking has been estimated from the stability constants.^{13,70a,71} ¹⁹⁵Pt NMR downfield shifts observed upon adduct formation between Pt(II) complexes such as [Pt(phen)(Arg)]²⁺ and mononucleotides indicated that the electron density of Pt(II) is decreased by stacking interaction, and this may be due to delocalization of electrons over the metal ion and the stacked π -systems.¹⁰⁵ The above interaction may be expressed as a metal-assisted stacking interaction between a coordinated aromatic ring (M π) and an uncoordinated aromatic ring of AA (L π), M π –L π . From this point of view the effects of metal assisted stacking interactions on the complex stability were studied for Cu(DA)(AA) with DA = bis(2-pyridylmethyl)amine (dpa), 2,2'-bipyrimidine (bpm),

and 4,4'-disubstituted 2,2'-bipyridines (Y₂bpy; Y = H, CH₃, N(Et)₂, COOEt, Cl) and AA = Trp, Tyr, and *p*-substituted phenylalanines (XPhe; X = H, NH₂, NO₂, halogens).¹⁰⁶ The X-ray crystal structure analysis of [Cu(dpa)(Trp)]ClO₄·2H₂O and [Cu((CONH₂)₂bpy)(Phe)]ClO₄·H₂O (**9**) (CONH₂)₂bpy = 4,4'-dicarbamoyl-2,2'-bipyridine) showed that the indole and benzene rings located close above the Cu(DA) coordination plane with the closest Cu(II)–C distances of 3.06 and 3.17 Å, respectively, which are shorter than the distances between the two rings, indicating that L π interacts with both the metal center and M π (Fig. 9). The structures show a strong distortion of the Cu(II) coordination plane, reflecting these interactions. Contribution of intramolecular stacking interactions to the stability of ternary metal complexes has been revealed by Sigel *et al.*, who calculated the stability enhancement from the stability constants by various methods.⁷¹ We evaluated the stabilization of complexes due to intramolecular interactions according to them and/or by considering a hypothetical equilibrium (eqn (1)) for formation of the complex species with intramolecular interactions from the species without them with Cu(en)(Ala) (en = ethylenediamine) as standard:^{70a,107}



where the interactions or M π –L π interactions are possible only in Cu(DA)(AA). The log*K* values are larger for AA = Trp and Tyr than for AA = Phe and are in a nearly linear relationship with the Hammett σ_p values of the substituents in AA for the systems with AA = NH₂Phe, Tyr, Phe, FPhe, and NO₂Phe (Fig. 10). The systems except DA = (NEt₂)₂bpy have negative slopes, which implies that M π with a lower electron density interacts with electron-rich L π more strongly than with electron-deficient L π . The metal-coordinated pyridine ring may be comparable with the *N*-methylpyridinium ring in the systems studied by Hunter *et al.*¹⁰⁸ and the perfluorobenzene ring in the systems by Cozzi and Siegel.¹⁰⁹ The results appear to be in contrast with the reported stability sequence for aromatic–aromatic interactions in the absence of a bound metal ion, π -deficient– π -deficient > π -deficient– π -rich > π -rich– π -rich.^{72a,72b,72d,109} These results suggest that the properties and reactivities of the central metal ion are modulated by vicinal aromatic rings (*vide infra*). An anomalously large stability enhancement observed for ClPhe and BrPhe may be due to the polarizability and van der Waals interactions of the halogen atoms. In this connection, Tanaka and Tabata¹¹⁰ calculated the stability enhancement due to ligand–ligand

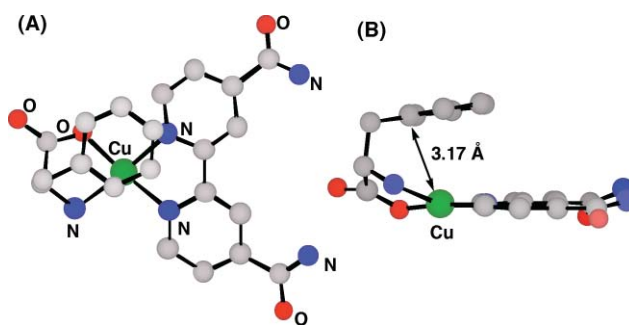


Fig. 9 Structure of [Cu((CONH₂)₂bpy)(Phe)]⁺ (**9**): (A) top view and (B) side view.¹⁰⁶

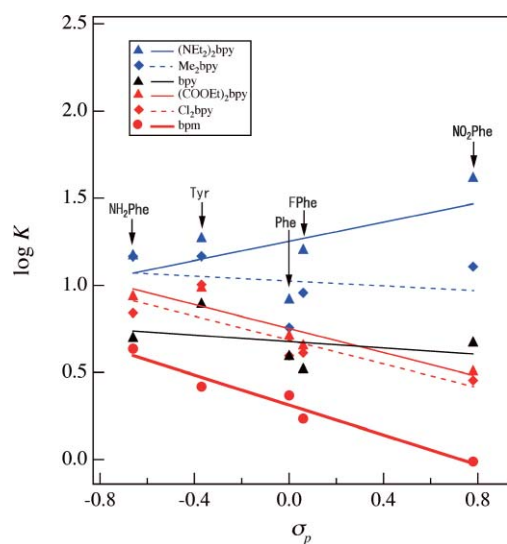


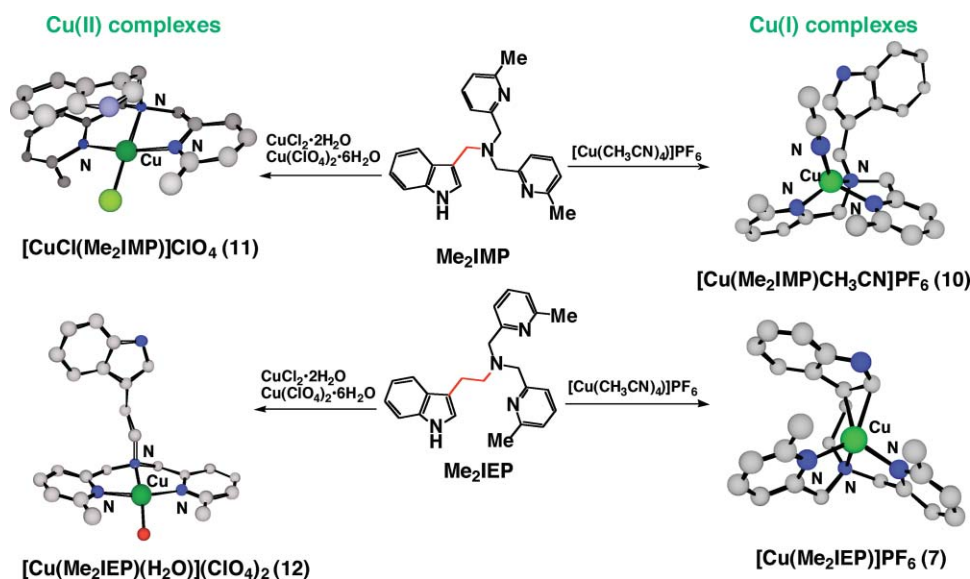
Fig. 10 Plots of $\log K$ values against the Hammett σ_p values of substituents X.¹⁰⁶

interactions according to the equation for the formation constants based on Tanaka's mechanistic consideration,¹¹¹ and from the plots of the enhanced stability against the free energy of transfer of the amino acid side chain from water to ethanol or dioxane, they discriminated between the contributions of π - π stacking and hydrophobic interactions. According to this method, the large stability enhancement for the haloaryl side chains is mainly due to hydrophobic interactions.

Indole-containing ligands, Me₂IMP and Me₂IEP having different lengths of the 3-indolylalkyl chain (a methylene and an ethylene arm for Me₂IMP and Me₂IEP, respectively) form Cu(I) and Cu(II) complexes, [Cu(Me₂IMP)(CH₃CN)]PF₆ (**10**), [CuCl(Me₂IMP)]ClO₄ (**11**), complex **7**, and [Cu(Me₂IEP)(H₂O)](ClO₄)₂ (**12**), respectively (Scheme 1).^{61,63} The complexes have structures with different conformations of the indole side chain depending on the length of the 3-indolylalkyl chain. While

there was no interaction between the indole ring and Cu(I) in complex **10** having a coordinated acetonitrile in the crystal structure, an indole-pyridine stacking interaction and an accompanying close Cu(II)-indole contact were observed for complex **11** with a 3N1Cl donor set. In the tetrahedral Cu(I) complex **7**, the indole ring was found to be bound to Cu(I) in η^2 fashion at the C(2)-C(3) moiety (Fig. 5(A)), but an extended conformation of the indole side chain without weak interactions was observed for the Cu(II) complex **12** with a 3N1O donor set.⁶³ These results suggest that the affinity of the indole ring for the Cu center depends on the valence state of Cu and the length of the indolylalkyl side chain. Similar bonding modes were reported for substituted phenyl and naphthyl rings.^{74,112} Itoh *et al.* studied structural and thermodynamic aspects of metal-aromatic ring interactions in Cu(I) complexes;⁷⁴ close contact between a side chain aromatic ring and the metal center, called d- π interaction, was detected for Cu(I) complexes containing a side chain phenyl group with the Cu-C distances of 2.17–2.66 Å. These Cu-phenyl bond lengths are comparable with or slightly longer than those revealed for complex **7** but may be of similar nature. They were found to influence the reactivity of the Cu(I) complexes with dioxygen; the stronger the interaction, the lower the reactivity.⁷⁴

At some type 1 Cu centers a close contact has been observed between an amino acid side chain alkyl or aromatic group and the trigonal Cu plane. A multi-copper oxidase, laccase from *Coprinus cinereus*, has the type 1 Cu site with a methyl group from the branched alkyl side chain of a leucine residue (Leu 462) above the Cu plane (the Cu...C distance: 3.51 Å) (Fig. 11(A)),¹¹³ where there is a methionine thioether sulfur atom as the fourth ligand in higher plant plastocyanins.^{48,114} A similar Cu...alkyl group contact was also disclosed for the type 1 Cu site in domain 2 of ceruloplasmin.¹¹⁵ For chemical systems, the Cu(II) complex of the *erythro* form of ephedrine derivatives was reported to have an axial contact between Cu(II) and the alkyl C-H moiety with the Cu...H distance of 2.454 Å.¹¹⁶ Structural and spectroscopic characterization of the metal...H-C interaction was made for the Cu(II) and Pd(II) complexes of the *N*-alkyl (R) derivatives of



Scheme 1 Behaviour of the side chain indole ring in Cu(I) and Cu(II) complexes.⁶³

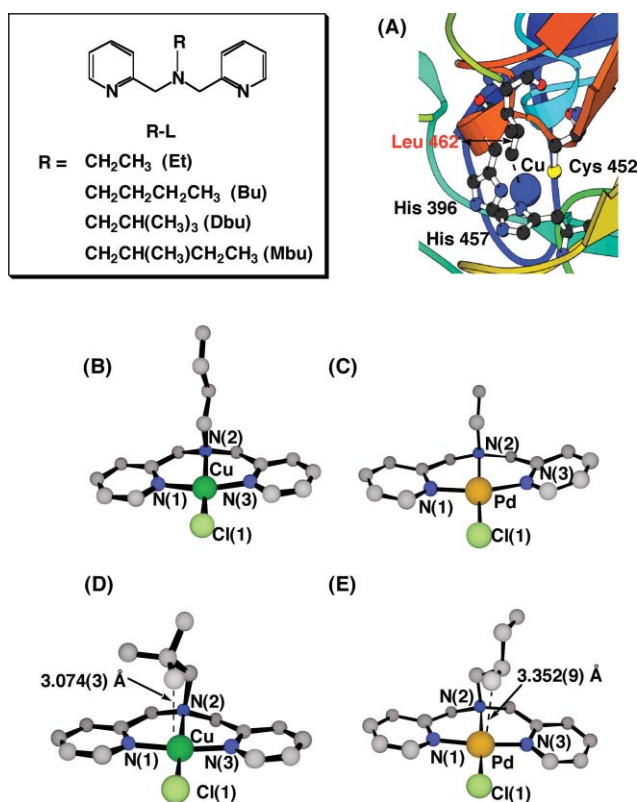


Fig. 11 Conformations of alkyl side chains in (A) *Coprinus cinereus* laccase¹¹³ and model complexes with a 3N1Cl donor set.¹¹⁷ (B) $[\text{CuCl}(\text{Bu-L})]^+$, (C) $[\text{PdCl}(\text{Et-L})]^+$, (D) $[\text{CuCl}(\text{Dmp-L})]^+$ (13) and (E) $[\text{PdCl}(\text{Mbu-L})]^+$ (14).

N,N-bis(2-pyridylmethyl)amine (R-L) (Fig. 11(B)–(E)).¹¹⁷ While the complexes with an ethyl or a butyl side chain, M-Et-L ($\text{M} = \text{Cu}(\text{II})$ and $\text{Pd}(\text{II})$) and $\text{Cu}(\text{II})\text{-Bu-L}$, respectively, showed no metal $\cdots \text{H-C}$ contact in the solid state, the complexes with a branched side chain, M-Dmp-L and M-Mbu-L ($\text{M} = \text{Cu}(\text{II})$ and $\text{Pd}(\text{II})$), were revealed to have very similar structures with a methyl group above the coordination plane with the $\text{M} \cdots \text{C}$ distances of 3.056(3)–3.304(6) and 3.204(3)–3.352(9) Å for $\text{M} = \text{Cu}(\text{II})$ and $\text{Pd}(\text{II})$, respectively, as illustrated in Fig. 11 for $[\text{CuCl}(\text{Dmp-L})]^+$ (13) and $[\text{PdCl}(\text{Mbu-L})]^+$ (14). The ^1H NMR spectra (in acetone- d_6) of the $\text{Pd}(\text{II})$ complexes with the $\text{Pd} \cdots \text{H-C}$ contact exhibited downfield shifts ($\Delta\delta = \delta_{\text{complex}} - \delta_{\text{ligand}}$) of 0.37–0.40 ppm of the methyl proton signals. The spectra indicated that the two methyl groups of complex 14 are distinguishable, while the three methyl groups of $\text{Pd}(\text{II})\text{-Dmp-L}$ are spectroscopically equivalent probably due to fast exchange of $\text{Pd} \cdots \text{CH}_3$ interaction. The finding supported that the interaction persists in solution to a considerable extent. Interestingly the $\Delta\delta$ values decreased with the increase of the dielectric constant of the solvent used, the order being acetone > methanol > acetonitrile > DMSO (Fig. 12). The $\text{Pd}(\text{II})$ complex with Et-L, $[\text{PdCl}(\text{Et-L})]$ (Fig. 11(C)), has the $\Delta\delta$ value of ~ 0.3 ppm in acetone (Fig. 12), indicating that the ethyl group also tends to be close to the $\text{Pd}(\text{II})$ center in less polar solvents. The $\text{Cu}(\text{II})$ complexes showed a d–d peak at 660–662 nm and quasi-reversible redox waves in methanol, but the redox potential of complex 13 was significantly higher and that of Cu-Mbu-L was slightly higher than the values for the complexes

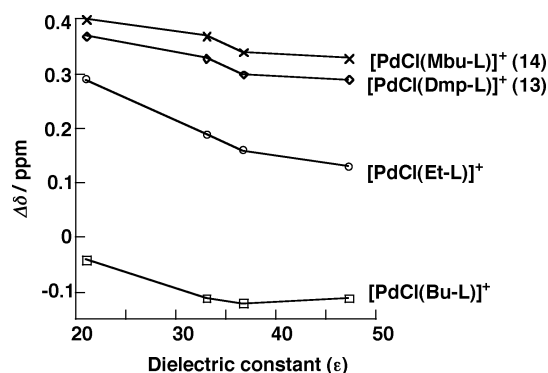


Fig. 12 Dependence of ^1H NMR downfield shifts, $\Delta\delta$, on the polarity of solvents.¹¹⁷

without the $\text{Cu} \cdots \text{H-C}$ contact. The solvent dependence of the $\Delta\delta$ values and the anodic shift of the redox potential suggest that the metal $\cdots \text{H-C}$ contact is of electrostatic nature and regarded as a weak hydrogen bond as pointed out by Thakur and Desiraju,¹¹⁸ and a theoretical study was also in line with this.¹¹⁹ The fact that both fungal laccases¹¹⁴ and ceruloplasmin¹²⁰ have an unusually high redox potential might indicate the effect of the observed $\text{Cu} \cdots \text{alkyl}$ group contact as a weak hydrogen bond. A recent study on the laccase from a fungus *Trametes trogii* showed that the type 1 Cu site has an isoleucine and a Phe residue sandwiching the trigonal Cu center.¹²¹ Since the ESR spectra for various laccases are very similar, the high redox potentials observed for fungal laccases was inferred to be due not only to the structural factor but also to other factors such as solvent accessibility and hydrogen bonding due to the second sphere residues.¹²¹

Imidazole and imidazolate complexes

The histidine imidazole group is involved in most of the metal binding sites of metalloenzymes particularly at the copper sites.^{4,34–36,48,122} The conjugate base imidazolate ion is present as a bridging ligand between copper and zinc ions in superoxide dismutase (CuZn-SOD) (Fig. 13).^{16,123} Lippard *et al.* carried out a series of pioneering work on SOD model complexes^{17,124} and designed and synthesized imidazolate-bridged dinuclear complexes as SOD mimics,¹²⁵ which provided valuable insights into the structure and function of the active site. Compared with $\text{Cu}(\text{II})\text{-Cu}(\text{II})$ homodinuclear complexes, only a small

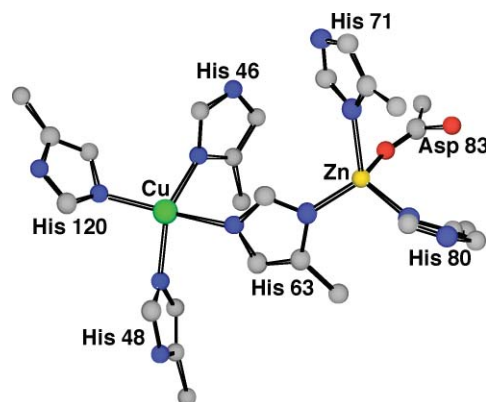


Fig. 13 Schematic representation of the active site of CuZn-SOD .¹⁶

number of imidazolate-bridged heterodinuclear CuZn-SOD model complexes have been reported.¹²⁶ Such complexes have been constructed by the self-assembly strategy using two independent mononuclear complexes and a bridging imidazole ring. We synthesized novel dinucleating ligands, Hbdpi and HMe₄bdpi (H denotes a dissociated proton in the imidazole ring) (Fig. 14), which reacted with Cu(ClO₄)₂·6H₂O and triethylamine at room temperature to give [Cu₂(bdpi)(CH₃CN)₂](ClO₄)₃·2CH₃CN·3H₂O (**15**) and [Cu₂(Me₄bdpi)(H₂O)₂](ClO₄)₃·4H₂O (**16**).^{127,128} When equimolar amounts of Cu(ClO₄)₂ and Zn(ClO₄)₂ were employed in the reaction with 1 equivalent of Hbdpi followed by addition of triethylamine, a heterodinuclear complex, [CuZn(bdpi)(CH₃CN)₂](ClO₄)₃·2CH₃CN (**17**), crystallized preferentially from CH₃CN.^{127,128} Without triethylamine, however, neither heterodinuclear nor homodinuclear complexes could be isolated. In the case of HMe₄bdpi, no heterodinuclear complex could be isolated, because the metal binding was not selective. The X-ray crystal structure revealed that the Cu(II)–Zn(II) heterodinuclear complex **17** has a structure which is essentially the same as that of the corresponding Cu(II)–Cu(II) homodinuclear complex **15**, as shown in Fig. 14. The SOD activities of complexes **15**, **16**,

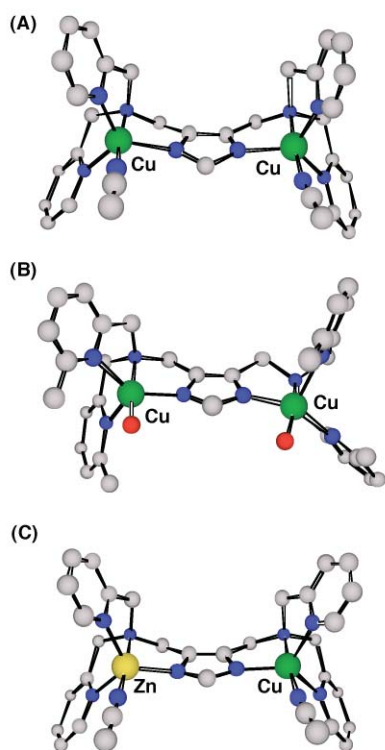
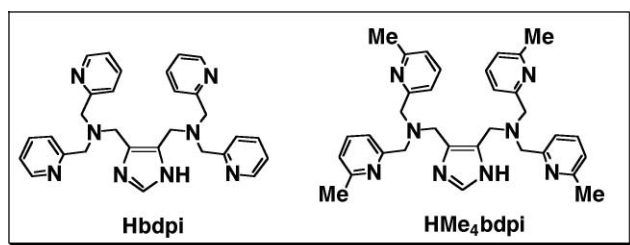


Fig. 14 Structures of imidazolate-bridged dinuclear complexes: (A) [Cu₂(bdpi)(CH₃CN)]³⁺ (**15**), (B) [Cu₂(Me₄bdpi)(H₂O)]³⁺ (**16**) and (C) [CuZn(bdpi)(CH₃CN)]³⁺ (**17**).^{127,128}

and **17** as measured by the concentrations of complexes required to attain 50% inhibition of the ferricytochrome *c* reduction (IC₅₀)¹²⁹ were found to increase in the order **16** < **15** < **17**. Although the IC₅₀ value of complex **17** (0.24 μM) is larger than the value reported for native SOD (0.04 μM), it has a coordination site available for the binding of superoxide (Fig. 14) and is among the most active SOD model complexes.¹²⁸

On the other hand, the chirality and side chain conformation of amino acid complexes have been used for construction of chiral assemblies,¹³⁰ and the influence of the stacked and unstacked forms of ternary Cu(II) complexes [Cu(DA)(AA)] on the helicity of polymeric structures was reported.¹³¹ Copper(II)-imidazole complexes are important as precursors in molecular architecture based on coordination chemistry to achieve highly ordered, symmetrical molecular or supramolecular structures.^{18,132} They can form cyclic molecules with a cavity, which may align and stack in the solid state to create semi-infinite channel structures potentially suitable as building blocks for microporous compounds.¹⁴ Deprotonation of Cu(II)–Schiff base complexes containing imidazole moieties was reported to give the corresponding tetramers or hexamers depending on the steric effect of the substituent of the imidazolate C(2) position.^{18,133} Recently, we synthesized and characterized the imidazolate-bridged tetra- or hexanuclear Cu(II) complexes of new tripodal polyimidazole ligands, Him-im₂ and Hha-im₂ (Fig. 15).¹³⁴ The reaction of Him-im₂ with Cu(ClO₄)₂·6H₂O under basic conditions gave a tetranuclear Cu(II) complex, [Cu₄(im-im₂)₄](ClO₄)₄ (**18**), whereas Hha-im₂ gave two polynuclear complexes, a tetramer [Cu₄(ha-im₂)₄](ClO₄)₄ (**19**) and a hexamer [Cu₆(ha-im₂)₆](ClO₄)₆ (**20**), exhibiting discretely different structures (Fig. 15 and 16, respectively).¹³⁴ The selectivity of formation of **19** or **20** is affected by a small difference in conditions such as the concentration of the reaction mixture, since the geometries of the Cu(II) center in **19** and **20** are similar. Complex **18** with a pseudo C₄ symmetry through the center of the molecule has a higher symmetry than **19**, which has a C₂ symmetry (Fig. 15). A plane consisting of four imidazolate-bridged Cu(II) ions in **18** gives a large space at the center of the

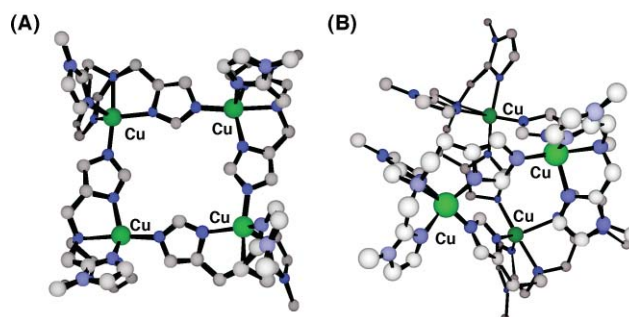
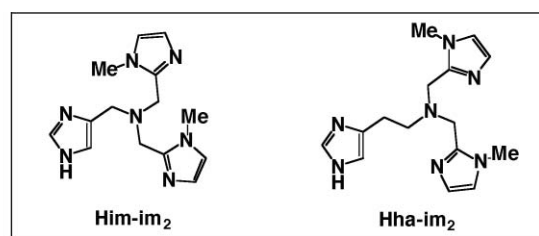


Fig. 15 Structures of imidazolate-bridged tetranuclear complexes: (A) [Cu₄(im-im₂)₄]⁴⁺ (**18**) and (B) [Cu₄(ha-im₂)₄]⁴⁺ (**19**).¹³⁴

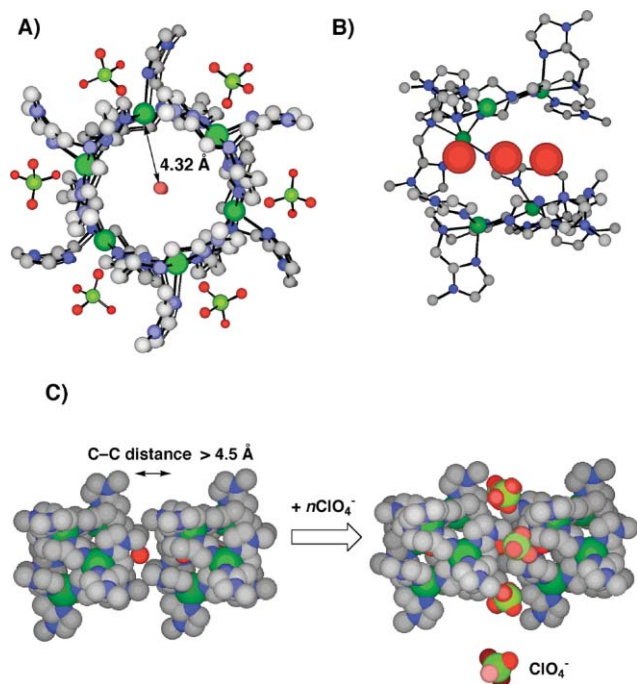


Fig. 16 Structure of the imidazole-bridged hexanuclear complex **20**: (A) top view, (B) view along the tube showing the one-dimensional water array and (C) side view of the tubular assembly in the solid state.¹³⁴

complex molecule, while the Cu(II) ions in **19** are positioned at the vertices of a distorted trigonal pyramid without forming a large space in the molecule. The coordination environments of the Cu(II) centers of **18** and **19** are different; the geometry of the Cu(II) center in **18** is distorted trigonal bipyramidal, whereas that in **19** is square-planar. This indicates that the structures of imidazole-bridged tetranuclear Cu(II) complexes are influenced by the coordination geometry of the Cu(II) site. In complex **20**, hexanuclear $[\text{Cu}_6(\text{ha-im}_2)_6]^{6+}$ ions which have an inner space with a diameter of *ca.* 8.5 Å are connected to form a tubular assembly by perchlorate ions and intermolecular interactions between the C(1) methyl protons or the C(5) proton of the imidazole and the oxygen atom of the perchlorate ion (Fig. 16). The tubes are aligned to the *c* axis and form ordered channel structures in the solid state. The side views of **20** (Fig. 16(B) and (C)) show the channel structure with some water molecules forming a straight line, and the distances between the imidazole C atoms and the O atom of the water molecule in the cavity exclude significant interactions between them (Fig. 16(B)).

Phenolate complexes and their oxidized species

Some metalloenzymes have coordinated phenolate or proximal phenol moieties of tyrosine residues at the active site, which are essential for the enzymatic activities.¹³⁵ Galactose oxidase is a single-copper oxidase (Fig. 17), and the active form has a Cu(II)–phenoxyl radical bond and catalyzes the conversion of primary alcohols to aldehydes in the presence of dioxygen.^{7–9,136} A number of model complexes of the galactose oxidase active site were reported, and detailed reviews on metal-phenoxyl radical complexes have been published.^{10,11,137–139}

The electronic structure of one-electron oxidized M(II)–phenolate complexes (M = Co, Ni, Zn, Pd, Pt) depends on the

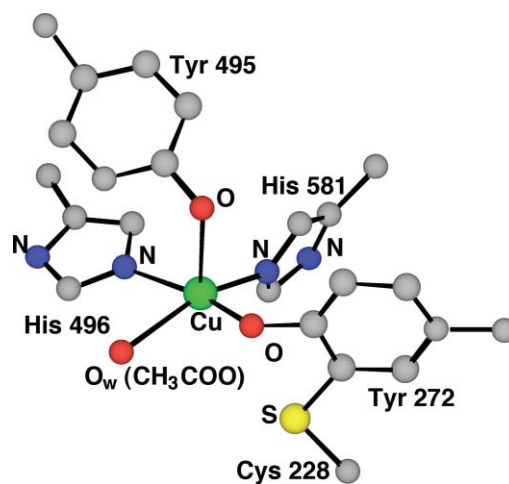


Fig. 17 Schematic representation of the active site of galactose oxidase (GO).⁷

metal ions and conditions such as solvents, temperature, and possible ligands existing in solution.^{138–142} The Cu(II)–phenolate complexes using four 3N1O tripodal ligands with a phenolate moiety have a mononuclear structure with a square-pyramidal geometry (Fig. 18).¹⁴⁰ $[\text{CuCl}(\text{HL})]\text{ClO}_4$ (**21**) has two pyridine nitrogen atoms, one tertiary nitrogen, and a chloride ion in the equatorial plane and a weakly coordinating phenol moiety at an

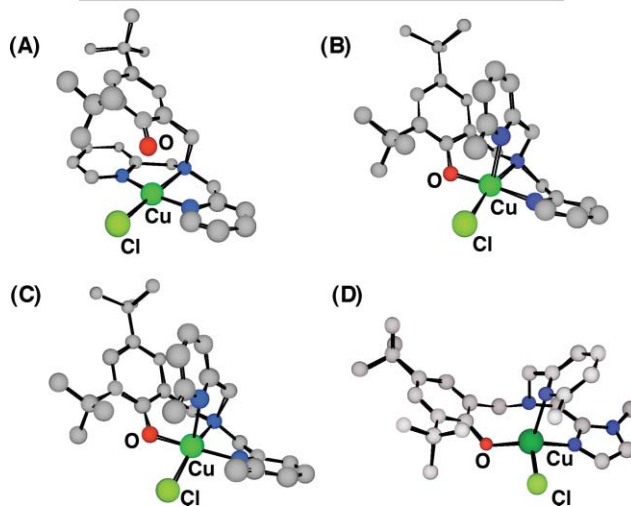
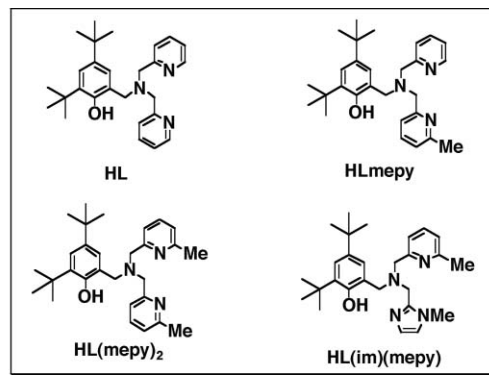
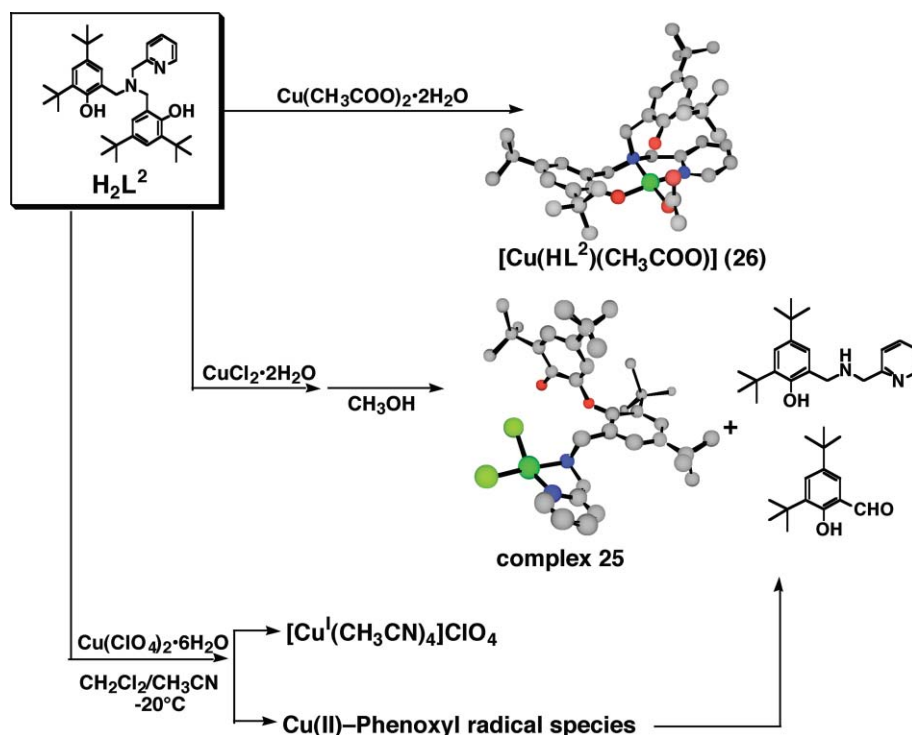


Fig. 18 Structures of Cu(II) complexes with phenol or phenolate moieties: (A) $[\text{CuCl}(\text{HL})]^+$ (**21**), (B) $[\text{CuCl}(\text{Lmepy})]$ (**22**), (C) $[\text{CuCl}(\text{L}(\text{mepy})_2)]$ (**23**) and (D) $[\text{CuCl}(\text{L}(\text{im})(\text{mepy}))]$ (**24**).¹⁴⁰

apical position, while the other three complexes, [CuCl(Lmepy)] (**22**), [CuCl(L(mepy)₂)] (**23**), and [CuCl(L(im)(mepy))] (**24**), have an equatorial phenolate coordination. In contrast to complex **21**, which showed an irreversible oxidation peak at 1.17 V, complexes **22**, **23**, and **24** showed a quasi-reversible redox wave around 0.6 V vs. Ag/AgCl, indicating that strong phenolate coordination at the equatorial position is required for stabilization of the one-electron oxidized form. The difference between **22**, **23**, and **24** lies in the stability of their phenoxyl radical species; the half lives of one-electron oxidized **22** and **24** were 65 and 62 min at −20 °C, respectively, but that of oxidized **23** was only 9.0 min at −40 °C. The stability order can be related with the nitrogen donor properties of the equatorial coordinating groups; complex **23** has a weaker nitrogen donation by the 6-methyl-2-pyridyl group than that of **22** and **24** for steric reasons, and therefore the oxidized form of **23** is less stable. Weaker donation by the 6-methyl-2-pyridyl group was confirmed by the stability constants for the Zn complexes of similar ligands without the *tert*-butyl groups of the phenolate moiety.¹⁴¹ A Cu(II)-phenoxyl radical species was prepared by the reaction of Cu(ClO₄)₂·6H₂O with the 2N2O-tripodal ligand containing two phenol moieties, H₂L², in CH₃CN/CH₂Cl₂ *via* disproportionation (Scheme 2).¹⁴³ The structures of the Cu(II) complexes of H₂L² and spontaneous formation of the radical species depended on the counter anions of the Cu(II) salts used: CuCl₂·2H₂O reacted with H₂L² to give a new Cu(II) complex **25** containing a bi(phenyl)ether moiety indicating formation of the phenoxyl radical species in the course of the reaction, whereas Cu(CH₃COO)₂·2H₂O gave [Cu(HL²)(CH₃COO)] (**26**), which is structurally very similar to the active site of galactose oxidase (Fig. 17) but did not give a detectable amount of the phenoxyl radical species (Scheme 2). A similar 2N2O-tripodal ligand, H₂MeOL², containing one *o*-methoxyphenol and one 2,4-di(*tert*-butyl)phenol moiety gave

the phenoxyl radical complex, [Cu₂(MeOL²)₂](ClO₄)₂ (**27**), as a reddish powder, which showed the activity to oxidize primary alcohols to aldehydes.¹⁴⁴ Such Cu(II)–phenoxyl radical complexes of similar 2N2O-tripodal ligands catalytically oxidized primary alcohols in the presence of a small amount of base.¹⁴⁵ On the other hand, a dianionic square-planar ligand containing two 2,4-di(*tert*-butyl)phenol moieties, H₂tbu-salcn (Fig. 19), was found to stabilize the Cu(III) oxidation state.¹⁴⁶ The locus of oxidation of [Cu(tbu-salcn)] (**28**) was metal-based, resulting in a Cu(III) species in the solid-state (Fig. 19(A)). In solution, a temperature-dependent valence tautomerism was observed between the high-valent metal ([Cu^{III}(tbu-salcn)]⁺) and ligand radical ([Cu^{II}(tbu-salcn)[•]]⁺) forms, indicating the near isoenergetic nature of these two species.

Comparative studies were made on the one-electron oxidized species of group 10 metal(II)–phenolate complexes. In a square-planar geometry, Ni(II) complexes have a low-spin d⁸ configuration similar to that of Pd(II) and Pt(II). Complexes of H₂tbu-salcn and H₂tbu-salcn including [Ni(tbu-salcn)] (**29**) (Fig. 19(B)), [Pd(tbu-salcn)] (**30**), and [Pt(tbu-salcn)] (**31**) have a mononuclear structure with two phenolate oxygen atoms coordinated in a very similar square-planar geometry.^{147,148} Cyclic voltammograms of the complexes showed a quasi-reversible redox wave at *E*_{1/2} = 0.82–1.05 V (vs. Ag/AgCl) corresponding to formation of the relatively stable one-electron oxidized species, and the molecular structures have been determined for the one-electron oxidized complexes isolated as crystals as shown for the Ni(II) complex **29** in Fig. 19(B).^{148–150} Upon one-electron oxidation, **29** was converted to the phenoxyl radical in CH₂Cl₂ and to the Ni(III)-phenolate species in DMF.¹⁴⁷ Conversion of the phenoxyl radical species to the metal centered oxidized species was observed upon addition of some exogenous ligand to the CH₂Cl₂ solution.¹⁵¹ The one-electron



Scheme 2 Reaction scheme for Cu(II)–H₂L² system showing the counterion dependence.¹⁴³

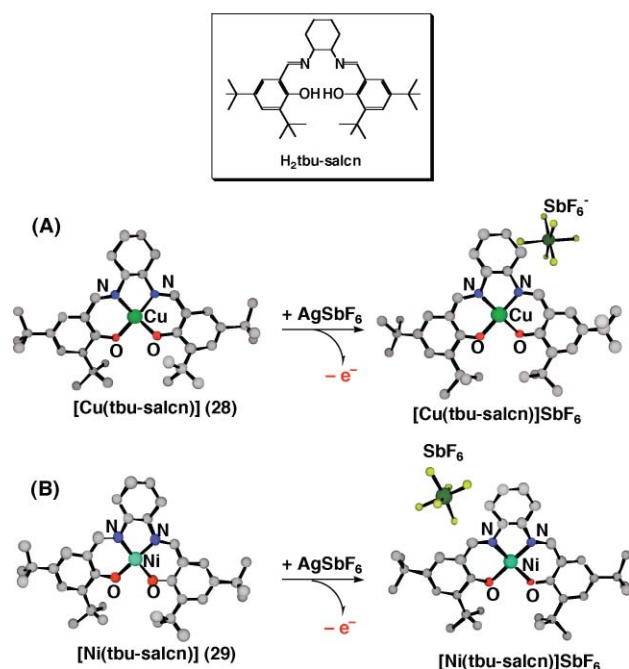


Fig. 19 Structures of Cu- and Ni-salen complexes, $[M(tbu-salcn)]$, before and after one-electron oxidation: (A) $[Cu(tbu-salcn)]$ (28)¹⁴⁶ and (B) $[Ni(tbu-salcn)]$ (29).¹⁵⁰

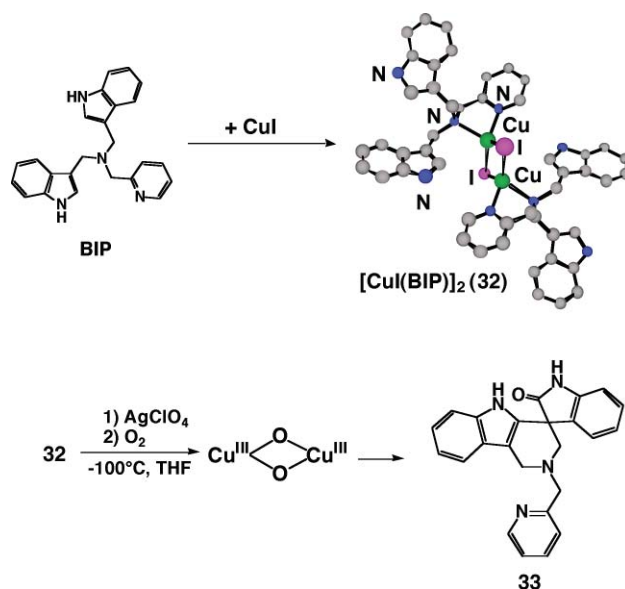
oxidized species of the Pd(II) complex **30** was different from that of the Ni(II) complex, the Pd(II)-phenoxyl radical species being the ground state in CH_2Cl_2 and in the solid sample in the range 5–300 K.¹⁴⁸ The ESR spectrum of the one-electron oxidized species of the Pt(II) complex **31** exhibited a temperature independent large g anisotropy in CH_2Cl_2 below $-80^\circ C$ and, together with the resonance Raman spectrum at $-60^\circ C$, indicated that the ground state of the Pt(II)-phenoxyl radical species has a large distribution of the radical electron spin at the Pt center. For the one-electron oxidized species of **31** isolated as crystals, $[Pt(tbu-salen)]SbF_6$, the oxidation state of Pt center was determined to be *ca.* +2.5 from the XPS and XANES measurements. Thus, the properties of the group 10 metal complexes with the 2N2O-donor *tbu-salen* coordination sphere are similar to each other and yet characteristic of the central metal ions.^{148,149}

Metal complexes with indole coordination and radical formation

Among the α -amino acids Trp has the largest aromatic ring, indole, in the side chain and thus forms a hydrophobic environment¹⁵² for specific binding of molecules and enzymatic functions⁵ as seen in galactose oxidase,⁷⁷ cytochrome *c* peroxidase,^{79,153} and *Anabena* flavodoxin.⁸² The indole ring has a pyrrole nitrogen, which is potentially available for coordination to the metal ion, although no metal-indole σ -bonding is known to date for biological systems (*vide supra*). Trp coordinates to the metal ions through the amino and carboxylate groups.^{3,4,13} The indole ring in ternary complexes with Cu(DA) such as $[Cu(DA)(Trp)]$ (DA = phen, bpy) is located above the coordination plane to be involved in the intramolecular π - π ($M\pi$ - $L\pi$) stacking interactions^{13,70a,71,106,154} and C(II)- $L\pi$ interactions^{3,4,70a} as detected in $[Cu(Tyr)_2]$ ¹⁵⁵ and Cu(GlyTrp) (GlyTrp = glycyltryptophan).⁶⁶ A

number of examples of complexes with direct metal-indole bonds have been reported.^{13,61}

Formation of the indolyl radical from a Trp residue in proteins has been established for the intermediate, compound I, formed in the catalytic reaction of cytochrome *c* peroxidase.^{79,153} Indolyl radical formation is possible with other biological systems, and Gray *et al.* reported the formation of the photogenerated tryptophan radical in modified azurins.¹⁵⁶ The Cu(I) complexes of the indole-containing ligands exhibited different reactivities with dioxygen. Formation of a bis(μ -oxo)dicopper(III) species as a short-lived intermediate could be observed in the reaction with O_2 of the Cu(I) complex of a ligand containing two indole-3-methylamine moieties (BIP), $[CuI(BIP)]_2$ (**32**) (Scheme 3).¹⁵⁷ This reaction gave a bis(indole) species **33** as one of the final products, which may indicate that an indole-radical species was formed as intermediate in the course of the reaction. Characterization of the indole radical in metal complexes was made with the use of the Pd(II) complexes of various 2N1O-donor ligands containing a pendent indole ring.^{158,159} The reactions of $PdCl_2$ with Htbu-iepp in CH_2Cl_2/CH_3CN at room temperature and in CH_3CN under refluxing gave an orange complex, $[PdCl(tbu-iepp)]$ (**34**), and a yellow complex with the same formula, $[PdCl(tbu-iepp-c)]$ (**35**), respectively (Fig. 20).^{158,159} Complex **35** was revealed to have the indole C(2) atom bound to Pd(II) with the distance of 1.973(2) Å. The redox potential of the indole ring in **35** was found to be lower than that in **34**, showing that the Pd(II)-bound indole moiety is more easily oxidized than the unbound one. Actually, oxidation of **35** with 1 equivalent of Ce(IV) in DMF gave the corresponding one-electron oxidized species, which exhibited an ESR signal at $g = 2.004$ and an absorption peak at ~ 550 nm corresponding to the formation of the Pd(II)-indole π -cation radical species.¹⁵⁸ Recently, a very similar indole- π -cation radical species was also observed in the course of the one-electron oxidation of the Pt(II)-indole binding complex, $[PtCl(tbu-iepp-c)]$.¹⁶⁰



Scheme 3 Structure of a Cu(I) complex containing two indole rings, $[CuI(BIP)]_2$ (**32**), and its reaction with O_2 .¹⁵⁷

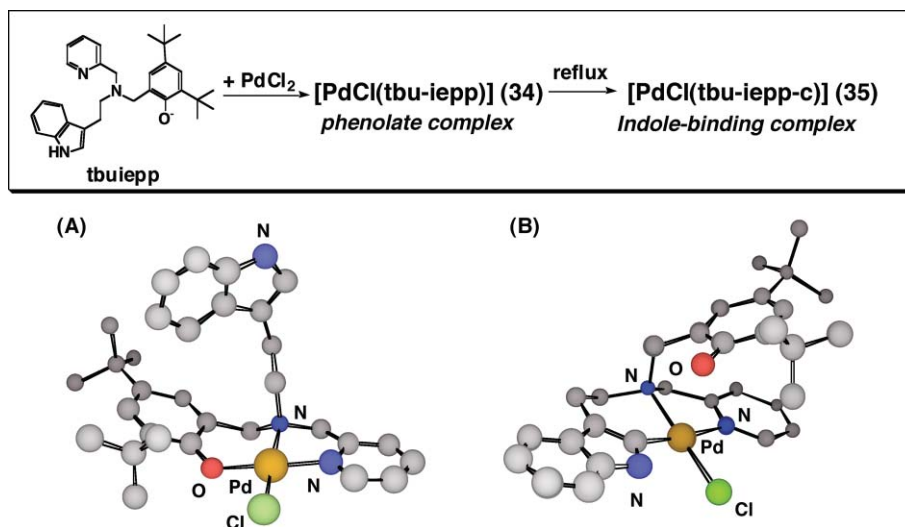


Fig. 20 Structures of Pd(II)-tbu-iepp complexes: (A) phenolate complex $[\text{PdCl}(\text{tbu-iepp})]$ (**34**) and (B) indole-binding complex $[\text{PdCl}(\text{tbu-iepp-c})]$ (**35**).¹⁵⁸

Concluding remarks

Amino acids are by far the most important low-molecular-weight ligands in biological systems. While the basic modes of their complex formation are rather simple and well known, the side chain groups participate in a wide variety of functions including metal bindings, weak interactions, and co-factor activities. In chemical systems the side chain groups are useful, for example, for construction of the artificial functionalized complexes as seen in the active site model complexes of metalloenzymes and the imidazolate-bridged multinuclear complexes forming molecular assemblies. Novel metal binding abilities have been revealed for the guanidine moiety of Arg and the indole ring of Trp in the respective proteins. The indole ring in a Cu(I) complex was found to bind with Cu(I) in η^2 fashion, while that in other complexes affected the reactivity with dioxygen possibly due to a field effect. Cation- π type interactions of Trp indole are also reported for Ca(II) and other metal ions. In the Cu(II)-binding octarepeat region of the prion protein³⁷ two Trp residues are reported to be close to the Cu-coordinated His imidazole¹⁶¹ and necessary for reduction of Cu(II) to Cu(I).^{161,162} The redox activities of the indole ring in complexes^{157–160} may suggest formation of the indole radical as seen in cytochrome *c* peroxidase^{79,153} and modified azurins¹⁵⁶ in the course of the Cu(II) reduction. The $\text{M(II)} \cdots \text{alkyl}$ group contact, which may be regarded as a weak hydrogen bond has been shown for Cu(II) and Pd(II) complexes with an alkyl side chain. The phenol moiety of the Tyr residue can act both as a ligand and a redox-active co-factor in biological systems, and the metal-phenoxyl radical complexes have been extensively studied for unravelling the tautomerism of the oxidation states and the mechanism of primary alcohol oxidation by galactose oxidase and model complexes.

In this perspective we focused on recent studies on weak interactions, unique metal binding abilities, and reactivities of some amino acid side chain groups and their possible biological relevance. These findings illustrate the roles of side chain groups surrounding the metal sites and suggest their further functional possibilities in proteins yet to be perceived.

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